# The Geometry of Linear Equations

One system $Ax = b$, three ways to see it: the row picture, the column picture, and the matrix form.

## PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.


1. **The problem: $Ax = b$**
   Turn a money question into equations. Three ways to read the same equations.
2. **Vectors and combinations**
   A fund as a list of numbers (an arrow). Scaling, adding, and mixing arrows.
3. **Matrix form**
   Write the whole system as one line, $Ax = b$, and see what $Ax$ means: multiply by columns or by rows.
4. **The row picture**
   Each rule is a line; the answer is where the lines cross.
5. **The column picture ★**
   Each fund is an arrow; the answer is the mix of arrows that lands on the target.
6. **Which targets can we reach?**
   All mixes of the columns. When do they cover every possible target?
7. **Three funds: 3 × 3**
   Add a third fund and a third rule. Planes in the row picture, 3D arrows in the column picture.
8. **When it goes wrong: singular**
   Funds that add nothing new: no answer, or endless answers. What changes with many funds.


METHODS IN THIS LESSON: Graphical method (row picture) · Column picture · Matrix–vector multiplication · Substitution

## PART 2 Prerequisites

**None. This is the first lesson.** You only need school arithmetic: adding, multiplying, negative numbers, and plotting a point $(x, y)$ on graph paper. Every new word is explained the first time it

appears; hover over (or tap) a dotted word to see its meaning.

Later lessons will link back here, for example to the column picture or matrix × vector, so you can jump to exactly the idea you need and return.

PART 3 **Explanation**

THE RUNNING EXAMPLE: A PORTFOLIO MIX

You have **₹10 lakh** to invest and two funds to choose from:

| Fund | Expected return per year | Return on ₹1 lakh |
|---|---|---|
| **Nifty 50 index fund** | 12% | ₹12 thousand |
| **Government bond fund** | 7% | ₹7 thousand |

You want to invest **all** ₹10 lakh and earn **₹1,05,000 a year** (₹105 thousand, a 10.5% return). **How much goes into each fund?**

The returns are illustrative round numbers, not forecasts.

## 1. The problem: $Ax = b$

Give the two amounts letters. These are the unknowns:

- $x$ = ₹ lakh put in the Nifty fund,
- $y$ = ₹ lakh put in the bond fund.

Now write each requirement as an equation:

- **Budget rule.** The two amounts must add up to ₹10 lakh: $x + y = 10$.
- **Return rule.** Each lakh in Nifty earns ₹12 thousand and each lakh in bonds earns ₹7 thousand, and together they must earn ₹105 thousand: $12x + 7y = 105$.

$$\begin{aligned} x + y &= 10 \quad (\text{budget, ₹ lakh}) \\ 12x + 7y &= 105 \quad (\text{return, ₹ thousand}) \end{aligned}$$

Two equations that must both be true at once are a system of equations. A solution is a pair $(x, y)$ that passes **both** rules.

The numbers that multiply the unknowns (the $1, 1$ and $12, 7$) are coefficients.

**DEFINITION: LINEAR EQUATION**

An equation is **linear** when each unknown is only multiplied by a number and the pieces are added. No squares, no unknowns multiplied together, no $\sqrt{x}$ or $\sin x$.

| Linear ✓ | Not linear ✕ |
|---|---|
| $x + y = 10$ | $x^2 + y = 10$ (a square) |
| $12x + 7y = 105$ | $xy = 21$ (unknowns multiplied together) |

"Linear" has the word *line* in it: a linear equation in two unknowns draws a straight line.

The **fundamental problem of linear algebra** is to solve such a system: $n$ linear equations in $n$ unknowns. This lesson looks at one system in three ways.

| View | You look at... | The question becomes... |
|---|---|---|
| **Row picture** | one equation (one rule) at a time | Where do the lines **meet**? |
| **Column picture** ★ | one unknown (one fund) at a time | What **mix** of the fund arrows makes the target? |
| **Matrix form** | the whole system at once | Find $x$ with $Ax = b$. |

The star marks the column picture: it is the view the rest of the course is built on.

▼ **Check yourself: write the system if the target return were ₹1,00,000.**

Only the right-hand side of the return rule changes: $x + y = 10$ and $12x + 7y = 100$.

## 2. Vectors and combinations

Look at what **₹1 lakh in the Nifty fund** does to our two rules. It uses ₹1 lakh of the budget and earns ₹12 thousand of return. Write those two numbers as a column:

$$\text{Nifty fund: } \begin{bmatrix} 1 \\ 12 \end{bmatrix} \qquad \text{Bond fund: } \begin{bmatrix} 1 \\ 7 \end{bmatrix} \qquad \leftarrow \begin{matrix} \text{budget used (₹ lakh)} \\ \text{return earned (₹ thousand)} \end{matrix}$$

**DEFINITION: VECTOR**

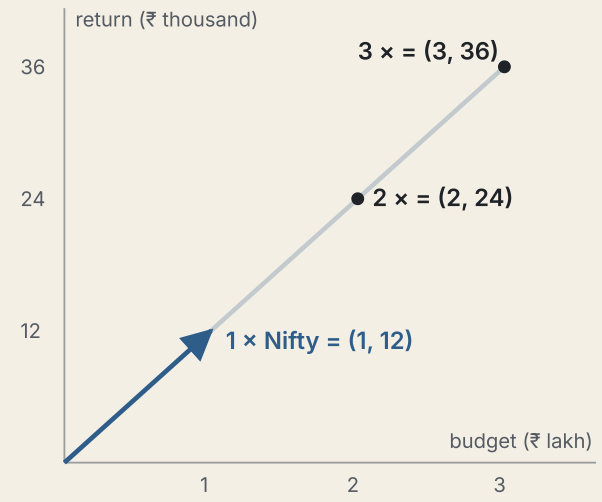
A **vector** is an ordered list of numbers, written as a column. Each number is a **component**. The Nifty vector $(1, 12)$ has components $1$ and $12$, and the order matters: first budget, then return.

A vector with 2 components can be drawn as an **arrow** from the origin: here, 1 step along the budget axis and 12 steps up the return axis.

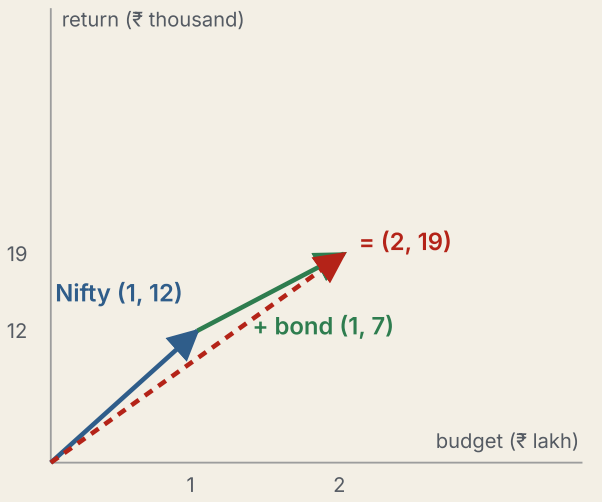
The set of **all** vectors with 2 components is called $\mathbb{R}^2$ (say "R-two"). $\mathbb{R}$ means the real numbers, and the little 2 says how many components. Likewise $\mathbb{R}^3$ is all lists of 3 numbers (3D space) and $\mathbb{R}^n$ is all lists of $n$ numbers, which we can't draw once $n > 3$ but can still calculate with.

### Two moves: scale and add

- **Scalar multiplication.** Multiply every component by one number (a scalar). ₹7 lakh in Nifty is 7 copies of ₹1 lakh: $7\begin{bmatrix}1\\12\end{bmatrix} = \begin{bmatrix}7\\84\end{bmatrix}$. Same direction, 7 times as long.
- **Vector addition.** Add matching components. ₹1 lakh in each fund: $\begin{bmatrix}1\\12\end{bmatrix} + \begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}2\\19\end{bmatrix}$: ₹2 lakh used, ₹19 thousand earned. As arrows: walk the first arrow, then walk the second from where you stopped.




**Scaling.** Every multiple of the Nifty arrow lies on one line through the origin: more money, same direction.




**Adding.** Walk the Nifty arrow, then the bond arrow from its tip. You end at $(2, 19)$: ₹2 lakh used, ₹19 thousand earned.

**DEFINITION: LINEAR COMBINATION**

Scale some vectors, then add them. For vectors $c, d$ and numbers $x, y$, the vector $x\,c + y\,d$ is a **linear combination** of $c$ and $d$. The numbers $x, y$ are its **weights**.

In our example a linear combination of the two fund vectors is simply **a portfolio**: $x$ lakh of Nifty plus $y$ lakh of bonds. Strang calls this "the most fundamental operation in the whole course".

▼ **Check yourself: what vector describes ₹4 lakh in Nifty plus ₹2 lakh in bonds?**

$4\begin{bmatrix}1\\12\end{bmatrix} + 2\begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}4\\48\end{bmatrix} + \begin{bmatrix}2\\14\end{bmatrix} = \begin{bmatrix}6\\62\end{bmatrix}$: ₹6 lakh used, ₹62 thousand earned.

## 3. Matrix form

Put all the coefficients in one table. A matrix is just a rectangular table of numbers:

$$\underbrace{\begin{bmatrix}1 & 1\\12 & 7\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x\\y\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}10\\105\end{bmatrix}}_{b} \qquad \text{or simply} \qquad Ax = b.$$

| Symbol | Name | In our example |
|---|---|---|
| $A$ | **coefficient matrix** | **row** $i$ = rule $i$ (budget, return); **column** $j$ = fund $j$ (Nifty, bond) |
| $x$ | vector of unknowns | the amounts $(x, y)$ in ₹ lakh |
| $b$ | **right-hand side** | the targets $(10, 105)$ |

A matrix with $m$ rows and $n$ columns is called $m \times n$ ("$m$ by $n$"). Ours is $2 \times 2$: 2 rules, 2 funds. Notice the columns of $A$ are exactly the fund vectors from topic 2.

**WATCH OUT**

The letter $x$ does two jobs: the single unknown $x$ (Nifty amount) and the whole vector of unknowns $x = (x, y)$. Strang writes the vector in bold. From the context it is always clear which one is meant.

## What $Ax$ means: matrix × vector, two ways

METHOD · MATRIX–VECTOR MULTIPLICATION

**By columns:** $Ax$ is a linear combination of the columns of $A$, with the entries of $x$ as weights. **By rows:** each entry of $Ax$ is the **dot product** of one row of $A$ with $x$. Same answer either way.

What does $Ax$ actually mean? Take our 2 × 2 matrix and the portfolio $x = (7, 3)$. There are two ways to multiply, and both give the same answer.

**Way 1: by columns (Strang's favourite).**

$$\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 7+3 \\ 84+21 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}.$$

Read it as "7 lakh of fund 1 plus 3 lakh of fund 2". This is the idea behind the column picture (topic 5).

**Way 2: by rows (dot products).** Multiply each row of $A$ by $x$, entry by entry, and add. This is a dot product:

$$\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 1 \cdot 3 \\ 12 \cdot 7 + 7 \cdot 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}.$$

Each row answers one rule: row 1 · $x$ = budget used, row 2 · $x$ = return earned. This is the idea behind the row picture (topic 4).

KEY FACT

**$Ax$ is a combination of the columns of $A$**, with the entries of $x$ as the weights. So "$Ax = b$ has a solution" means exactly "$b$ is a combination of the columns".

SIZE RULE

An $m \times n$ matrix times a vector with $n$ entries gives a vector with $m$ entries. The number of **columns** of $A$ must equal the number of entries of $x$: one weight per column (one amount per fund).

More generally $(m \times n)\,(n \times p) = (m \times p)$: the inner sizes must match, the outer sizes give the answer's shape.

▼ Check yourself: compute $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ both ways (Strang's example).

By columns: $1(2,1) + 2(5,3) = (2+10,\ 1+6) = (12,7)$. By rows: $2 \cdot 1 + 5 \cdot 2 = 12$ and $1 \cdot 1 + 3 \cdot 2 = 7$. Same answer, $(12,7)$.

▼ Check yourself: true or false? A 3 × 2 matrix times a 2 × 3 matrix is a 3 × 3 matrix.

**True.** The inner sizes (2 and 2) match, and the outer sizes give $3 \times 3$.

## 4. The row picture

METHOD · GRAPHICAL METHOD (ROW PICTURE)

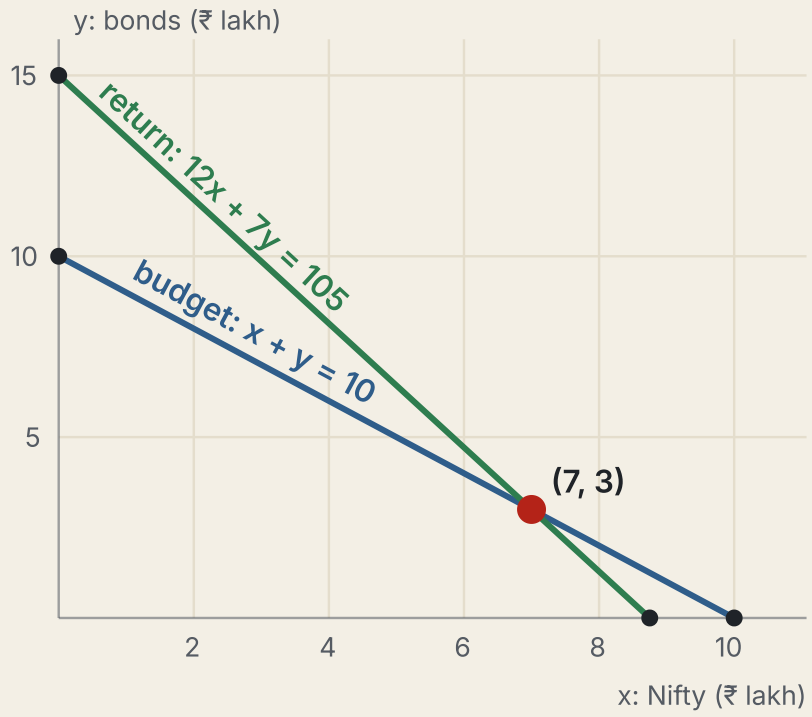
Draw each equation as a line (or a plane in 3D) and read off the point where they all meet. Great for seeing 2 unknowns; impractical beyond 3.

**Idea:** take one equation (one row) at a time and draw **every** point $(x, y)$ that satisfies it. A linear equation in two unknowns gives a straight line. A solution must satisfy both rules, so it sits on both lines: **it is where the lines cross.**

### Draw each line from two points

Two points fix a line. An easy method: set one unknown to $0$ and solve for the other.

| | **Budget:** $x + y = 10$ | **Return:** $12x + 7y = 105$ |
|---|---|---|
| Put $x = 0$ | $y = 10 \rightarrow$ point $(0, 10)$ | $7y = 105, y = 15 \rightarrow$ point $(0, 15)$ |
| Put $y = 0$ | $x = 10 \rightarrow$ point $(10, 0)$ | $12x = 105, x = 8.75 \rightarrow$ point $(8.75, 0)$ |
| Meaning | all-bonds or all-Nifty, both use ₹10 lakh | ₹15 lakh in bonds, or ₹8.75 lakh in Nifty, each earns ₹105 thousand |




**Row picture.** Each rule is a line. The only point on both lines is $(7, 3)$: ₹7 lakh in Nifty, ₹3 lakh in bonds.

**Check** the crossing point in both rules. Reading a point off a picture is not proof; the arithmetic is.

- Budget: $7 + 3 = 10$ ✓
- Return: $12 \times 7 + 7 \times 3 = 84 + 21 = 105$ ✓

So the answer is **₹7 lakh in the Nifty fund and ₹3 lakh in the bond fund.**

> **TIP: DOES THE LINE PASS THROUGH THE ORIGIN?**
>
> Strang always asks this first. A line passes through $(0, 0)$ **exactly when its right-hand side is** $0$. Neither of our lines does: investing nothing can't use ₹10 lakh or earn ₹105 thousand.

The row picture is the one you met in school, and it works well for two unknowns. With three unknowns each rule becomes a plane, and it gets hard to see (topic 7).

> **▼ Check yourself: where does the budget line cross the return line if the target return is ₹1,20,000?**
>
> Solve $x + y = 10$ and $12x + 7y = 120$. From the first, $y = 10 - x$. Then $12x + 70 - 7x = 120$, so $5x = 50$, $x = 10$, $y = 0$: all ₹10 lakh in Nifty. The lines cross on the $x$-axis at $(10, 0)$.

## 5. The column picture ★

**METHOD · COLUMN PICTURE (LINEAR COMBINATION OF COLUMNS)**

Write $Ax = b$ as $x\,(\text{column } 1) + y\,(\text{column } 2) + \cdots = b$ and look for the weights that land on $b$. The view the whole course builds on.

Now read the **same** system a column at a time. Pull out everything that multiplies $x$ and everything that multiplies $y$:

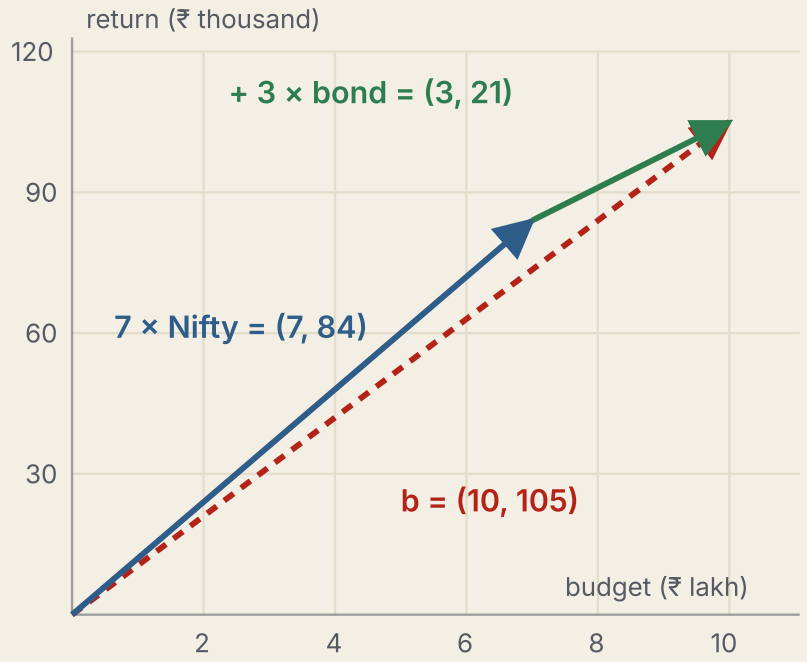
$$x\begin{bmatrix}1\\12\end{bmatrix} + y\begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}10\\105\end{bmatrix}$$

Read it aloud: *"$x$ lakh of Nifty plus $y$ lakh of bonds must give ₹10 lakh used and ₹105 thousand earned."* The left side is a linear combination of the columns of $A$. So the question has become:

**THE COLUMN QUESTION**

Which combination of the column vectors $(1, 12)$ and $(1, 7)$ produces the target vector $b = (10, 105)$?

We already know the answer from the row picture, $x = 7$ and $y = 3$, so draw that combination: 7 copies of the Nifty arrow, then 3 copies of the bond arrow from its tip.




**Column picture.** 7 Nifty arrows take you to $(7, 84)$; 3 bond arrows from there take you to $(10, 105)$, exactly the target $b$ (dashed).

Component by component: budget $7 + 3 = 10$, return $84 + 21 = 105$. Same answer as the row picture, seen a completely different way.

▼ **Check yourself: in the column picture, what does the point $(10, 70)$ mean, and which portfolio reaches it?**

₹10 lakh used and ₹70 thousand earned, a 7% return. Only bonds earn 7%, so $x = 0$, $y = 10$: $0 \cdot (1, 12) + 10 \cdot (1, 7) = (10, 70)$.

## 6. Which targets can we reach?

Strang now asks a bigger question: if we try **every** $x$ and **every** $y$, which right-hand sides $b$ can we produce? In money terms: which (budget, return) targets can some mix of these two funds hit?

**RESULT**

The combinations of $(1, 12)$ and $(1, 7)$ fill the **whole plane** $\mathbb{R}^2$. For **every** target $b$ there is exactly one mix $(x, y)$.

**Why:** the two arrows point in different directions. One arrow alone only reaches points on its own line (see the scaling figure in topic 2). A second arrow pointing a different way lets you move off that line, and with both you can reach anywhere.

▼ **Why it works: solve for any target $(B, R)$**

Take any budget $B$ and any return $R$. We need $x + y = B$ and $12x + 7y = R$.

From the first, $y = B - x$. Put that into the second: $12x + 7(B - x) = R$, so $5x + 7B = R$, giving

$$x = \frac{R - 7B}{5}, \qquad y = B - x = \frac{12B - R}{5}.$$

We only divided by $5 = 12 - 7$, which is not zero **because the two funds have different returns**. So a solution exists for every $(B, R)$, and it is the only one. If the returns were equal we would be dividing by zero: that is the singular case in topic 8.

**DEFINITION: SPAN**

The **span** of some vectors is the set of **every** point you can reach by mixing them: scale each one by any number (positive, negative or zero) and add. In portfolio words: every (budget, return) result you can get from these funds, in any amounts.

- **Vectors** = your available directions (here, the funds).
- **Span** = everywhere you can reach using those directions.

It is not about one target. It is about the **whole reachable area**: a line, a plane, or a whole space.

See how the span grows as we add funds:

| Funds you may use | Their span | What it means |
|---|---|---|
| Nifty only, $(1, 12)$ | a **line** through the origin: return $= 12\times$ budget | whatever you invest, you earn exactly 12% |
| Bonds only, $(1, 7)$ | a different **line**: return $= 7\times$ budget | you earn exactly 7% |
| Nifty and bonds | the **whole plane** $\mathbb{R}^2$ | any budget with any return (using negative amounts if needed) |

**DEFINITION: COLUMN SPACE**

The **column space** of a matrix $A$ is the span of its columns. Strang writes it $C(A)$. For us the columns are the funds, so $C(A)$ is **every target your funds can produce**.

For our matrix $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$, the column space is all of $\mathbb{R}^2$.

**WHY THE COLUMN SPACE MATTERS**

$Ax = b$ has a solution **exactly when $b$ is in the column space of $A$.**

The reason is one sentence: $Ax$ is always a combination of the columns of $A$ (see topic 3), so asking "is there an $x$ with $Ax = b$?" is the same as asking "is $b$ one of those combinations?".

- **Nifty + bonds:** $C(A)$ is the whole plane, so every target, including $(10, 105)$, has a solution.
- **Two Nifty funds** (topic 8): $C(A)$ is only the line return $= 12\times$ budget. The target $(10, 120)$ is on it (solvable); $(10, 105)$ is not (no solution).

**WATCH OUT: MATHS SAYS YES, THE MARKET MAY SAY NO**

"Every target" includes strange ones. Want a **15%** return on ₹10 lakh, ₹150 thousand? The formula gives $x = (150 - 70)/5 = 16$ and $y = (120 - 150)/5 = -6$. A negative amount means **borrowing** ₹6 lakh at the bond rate to put ₹16 lakh into Nifty. The algebra is happy; whether you are allowed to (or should) do that is a separate question.

▼ **Check yourself: which mix earns exactly 10% (₹100 thousand) on ₹10 lakh?**

$x = (100 - 70)/5 = 6$ and $y = (120 - 100)/5 = 4$. Check: $6 + 4 = 10$ ✓ and $72 + 28 = 100$ ✓.

## 7. Three funds, three rules: 3 × 3

Add a third fund and a third rule.

- **Midcap fund**: expected 15% a year (₹15 thousand per lakh), but riskier.
- **Market exposure rule.** A fund's beta says how much it moves when Nifty moves 1%. Nifty fund: beta $1.0$. Bond fund: beta $0$. Midcap fund: beta $1.2$. Each lakh of a fund adds "beta × 1 lakh" of Nifty-like exposure, and you want ₹7.4 lakh of exposure in total.

With $z$ = ₹ lakh in the midcap fund:

$$\begin{aligned} x + y + z &= 10 && (1)\ \text{budget} \\ 12x + 7y + 15z &= 111 && (2)\ \text{return} \\ 1.0x \quad + 1.2z &= 7.4 && (3)\ \text{market exposure} \end{aligned} \quad \Longleftrightarrow \quad \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}}_{b}$$

Now $A$ is $3 \times 3$. Each column is still one fund: (budget used, return earned, exposure added) per ₹1 lakh. The bond fund's column has a $0$ at the bottom because it adds no market exposure.

### Solve it step by step

> **METHOD · SUBSTITUTION**
>
> Solve one equation for one unknown, put that expression into the other equations, and repeat until one unknown is left. Fine by hand for small systems; Gaussian elimination (Lesson 2) is the systematic version every computer uses.

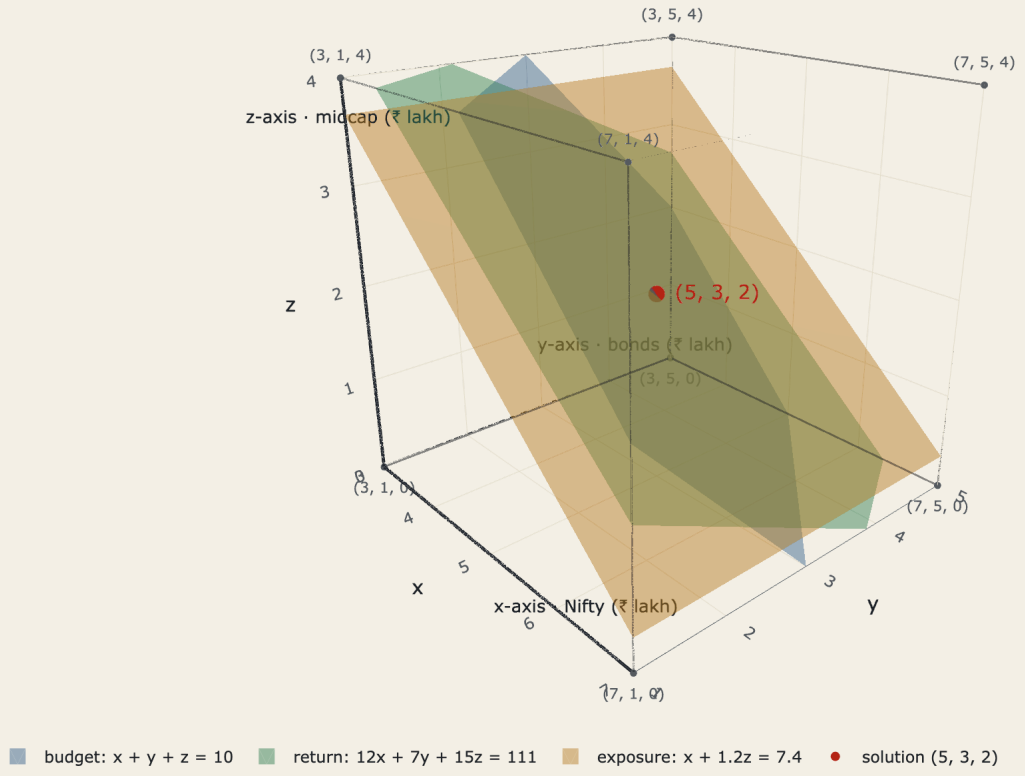
A systematic method, elimination, is Lesson 2. Here, substitution is enough: each step removes one unknown.

1. **Equation (3) has no $y$.** Write $x$ in terms of $z$: $x = 7.4 - 1.2z$.
2. **Equation (1) gives $y$.** $y = 10 - x - z = 10 - (7.4 - 1.2z) - z = 2.6 + 0.2z$.
3. **Put both into equation (2).** $12(7.4 - 1.2z) + 7(2.6 + 0.2z) + 15z = 111$. Multiply out: $88.8 - 14.4z + 18.2 + 1.4z + 15z = 111$, so $107 + 2z = 111$ and $z = 2$.
4. **Back-substitute.** $x = 7.4 - 2.4 = 5$ and $y = 2.6 + 0.4 = 3$.

**Check:** budget $5 + 3 + 2 = 10$ ✓; return $60 + 21 + 30 = 111$ ✓; exposure $5 + 0 + 2.4 = 7.4$ ✓. So **₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.**

### Row picture: three planes

With three unknowns, all the points $(x, y, z)$ satisfying one linear equation form a flat plane. Two planes usually meet in a line; the third plane cuts that line at one point, the solution.

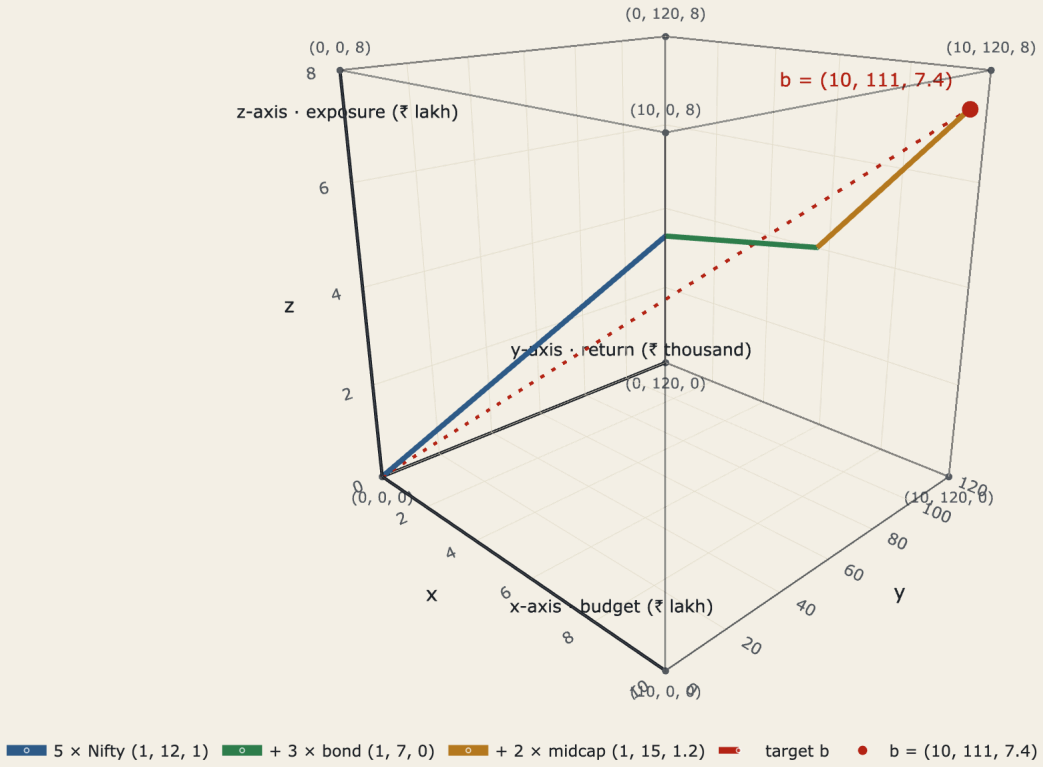



**Row picture (3 × 3), drag to rotate.** Budget, return and exposure planes all pass through $(5, 3, 2)$. Already harder to read than two crossing lines, and in four or more dimensions impossible to draw.

## Column picture: three arrows in 3D

$$x \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 15 \\ 1.2 \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}$$

Same question as before, one dimension up: which mix of the three fund arrows lands on $b$? Walk 5 Nifty arrows, then 3 bond arrows, then 2 midcap arrows.




**Column picture (3 × 3), drag to rotate.** $5(1, 12, 1) + 3(1, 7, 0) + 2(1, 15, 1.2) = (10, 111, 7.4)$, the target $b$.

## Change only the target

Keep the same three funds (same $A$) but change the goal to **₹98 thousand return and ₹5.2 lakh exposure**: $b = (10, 98, 5.2)$. The answer is $(4, 5, 1)$.

- Check: $4 + 5 + 1 = 10$ ✓; $48 + 35 + 15 = 98$ ✓; $4 + 0 + 1.2 = 5.2$ ✓.
- **Row picture:** three **new** planes (each right-hand side moved), now meeting at $(4, 5, 1)$.
- **Column picture:** the **same** three arrows, just mixed in different amounts. This is why Strang prefers columns: the arrows belong to $A$ and never change; only the target does.

> ▼ **Check yourself: which mix of the three funds gives $b = (1, 15, 1.2)$?**
>
> $b$ is exactly the midcap column, so $x = 0, y = 0, z = 1$: ₹1 lakh in midcap. Strang uses the same trick in the lecture (his right-hand side equals column 3).

## 8. When it goes wrong: singular matrices

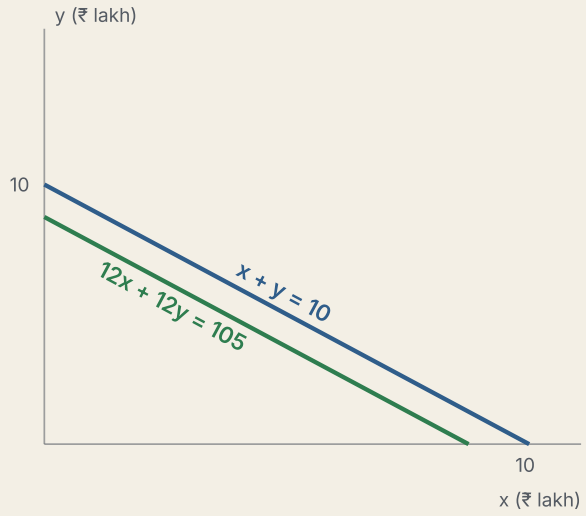
So far every target could be reached. Strang's key question:

> THE QUESTION, IN TWO LANGUAGES
>
> **Algebra:** can we solve $Ax = b$ for **every** right-hand side $b$?
>
> **Columns:** do the linear combinations of the columns fill the whole space?
>
> These are the same question.

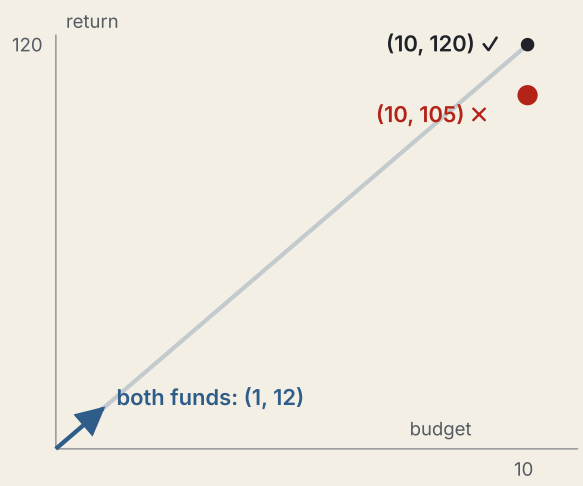
### 2 × 2: two funds that are really one

Replace the bond fund with a **second Nifty 50 index fund** from another company. It also earns 12%, so its column is also $(1, 12)$.

$$x + y = 10, \qquad 12x + 12y = 105.$$




**Row picture.** The lines are parallel (the second is $x + y = 8.75$). They never meet: **no solution**.




**Column picture.** Both arrows point the same way, so every mix stays on one line (return = 12 × budget). The target is off that line.

With these two funds you earn 12% whatever you do. If the target is ₹105 thousand there is **no solution.** If the target were exactly ₹120 thousand, **every** split works: the two lines would be the same line, giving **infinitely many solutions.**

**DEFINITION: SINGULAR AND INVERTIBLE**

A square matrix is **singular** when its columns fill **less** than the whole space, so some $b$ cannot be reached. Its columns are **dependent**: one column adds nothing new.

Otherwise it is **non-singular**, also called **invertible**: the columns fill the whole space and every $b$ has exactly one solution. Our Nifty + bond matrix is invertible; the two-Nifty matrix is singular.

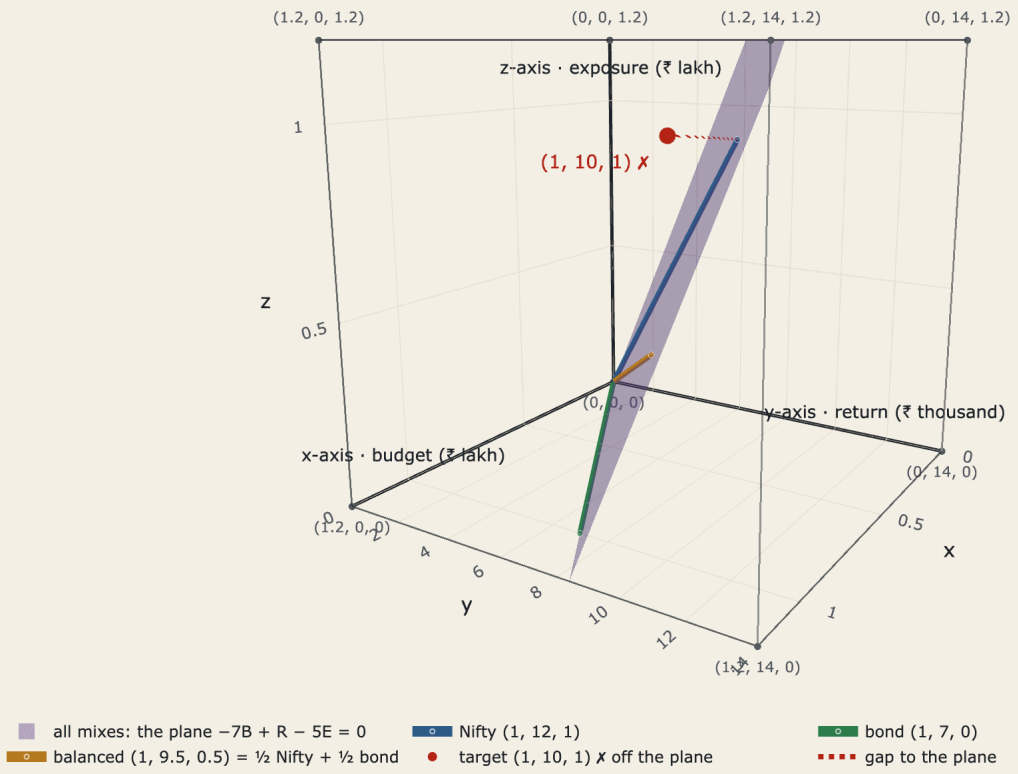
| What happens (2 × 2) | Row picture | Column picture | Number of solutions |
|---|---|---|---|
| Columns point different ways | lines cross once | arrows fill the plane | exactly one |
| Columns on one line, $b$ off it | parallel lines | $b$ off the arrows' line | none |
| Columns on one line, $b$ on it | the same line twice | $b$ on the arrows' line | infinitely many |

### 3 × 3: a fund that is just a mix of the others

Replace the midcap fund with a **balanced fund** that simply holds half Nifty, half bonds. Per lakh it uses ₹1 lakh, earns $\frac{1}{2}(12) + \frac{1}{2}(7) = 9.5$ thousand and adds $\frac{1}{2}(1) + \frac{1}{2}(0) = 0.5$ lakh of exposure:

$$\text{balanced} = \begin{bmatrix} 1 \\ 9.5 \\ 0.5 \end{bmatrix} = \tfrac{1}{2}\begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + \tfrac{1}{2}\begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix}.$$

The third column is a combination of the first two, so **all three columns lie in one plane**. Every portfolio you can build lies in that plane too, and any target off it is unreachable. This is exactly Strang's warning: "if these three columns all lie in the same plane, then their combinations will lie in that same plane".




**Singular 3 × 3, seen almost edge-on (drag to rotate).** The Nifty, bond and balanced arrows all lie flat in the shaded plane. The target $(1, 10, 1)$ sticks out of it: ₹1 lakh with ₹1 lakh of exposure must be all Nifty, which earns 12, not 10. No mix of these funds reaches it.

**WATCH OUT**

Three funds do not guarantee three independent directions. A fund built from the others adds no new direction, however different its name or factsheet looks.

**WITH MANY FUNDS**

Nothing changes with 9 funds and 9 rules: each column is a vector in $\mathbb{R}^9$ and the question is still "which mix of the columns reaches $b$?". You can't draw it, but the column picture still works. Funds picked at random almost always fill the whole space (invertible). Two copies of the same Nifty 50 index from different companies don't: the columns then fill only an 8-dimensional "plane" inside 9-dimensional space, and most targets are out of reach.

▼ **Check yourself: is the target $(10, 111, 7.4)$ reachable with Nifty, bond and balanced funds?**

Every reachable target $(B, R, E)$ satisfies $-7B + R - 5E = 0$ (the equation of the plane; each column passes, e.g. Nifty: $-7 + 12 - 5 = 0$). For $(10, 111, 7.4)$: $-70 + 111 - 37 = 4 \neq 0$. So **no**: it is off the plane.